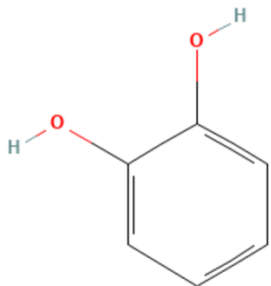


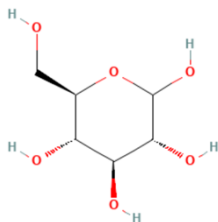
Representations of small molecules

Molecular Graphs

- A molecular graph represents molecules, with atoms represented as nodes and bonds represented as edges between nodes.



Catechol (C₆H₆O₂)

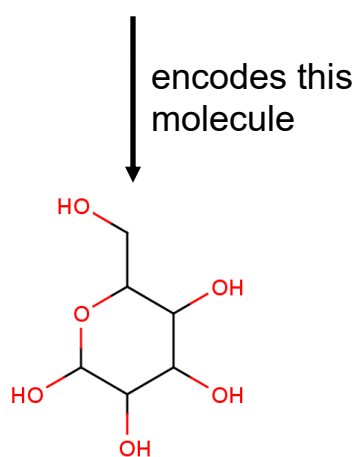


Glucose (C₆H₁₂O₆)

- Rules for Drawing Molecular Graphs:

- **Nodes:** Each node represents an atom (typically labeled with the element symbol)
Exception: sometimes carbon atoms do not have carbon symbol "C"
- **Bonds:** Each edge represents a bond between atoms
 - Number of parallel lines indicate type of bonds
- **Hydrogen on Carbon:** Hydrogens attached to carbon atoms are generally not shown.
- **Stereochemistry** (spatial arrangements of atoms):
 - Solid wedges: bonds coming out of the plane
 - Hashed wedges (hashed triangles): bonds going behind the plane.
 - Plain Lines: bonds in the plane of the screen

- SMILES - simplified molecular-input line-entry system
- A SMILES string represents the structure of a small molecule using a single string
 - Can consist of letters, numbers and special characters
- SMILES string for Glucose: C(C1C(C(C(C(O1)O)O)O)O)O



- Does not store stereochemical information yet, but we can add it

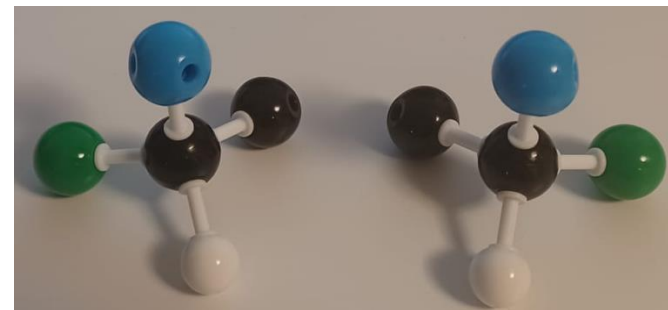
SMILES Specification Rules (1)

- Atoms are represented by their chemical symbols in square brackets, e.g. [Fe]
- Hydrogen atoms are usually implied and not explicitly written
- Brackets can be omitted if all of the following rules apply:

- The atom is part of the following atoms: C, N, O, P, S, F, Cl, Br, B, or I
- Has no formal charge
- Number of attached hydrogen atoms can be implied
- They are the normal isotopes
- They are not chiral centers

- Definition: A specific atom in a molecule that has four different groups attached to it, leading to molecules that are non-superimposable mirror images of each other.

[N-] ← Nitrogen with add. electron
[OH-]
[13C]

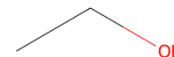


SMILES Specification Rules (2)

Bonds:

- Single bonds are not explicitly represented unless necessary ('-')
- Double and triple bonds are represented by '=' and '#'

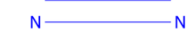
CCO



O=O



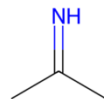
N#N



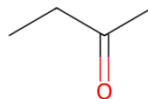
Branches

- described by using parentheses "(" and ")"

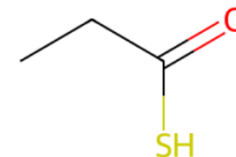
CC(=N)C



C(=O)(CC)C

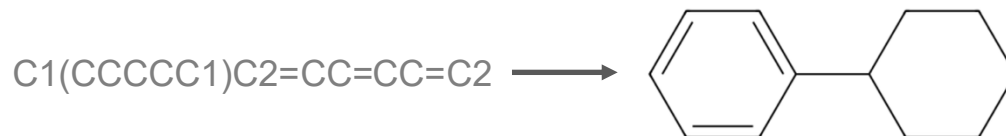
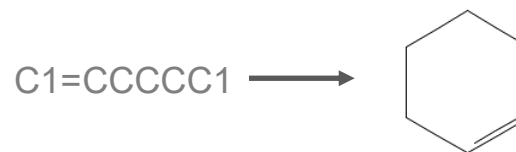
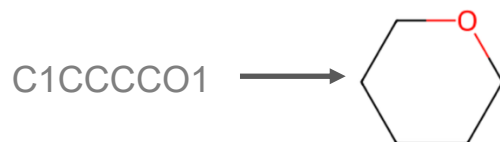


C(C(=O)S)C

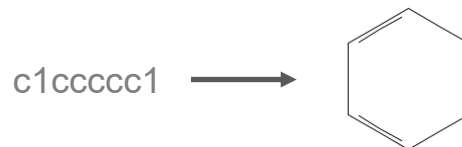


SMILES Specification Rules (3)

- Rings / cyclic structures are indicated by using numbers to represent the connection points. The same number is used for both ends of the ring.

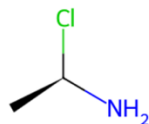


- Atoms that are part of an aromatic bond are denoted by lower case letters

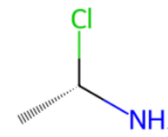
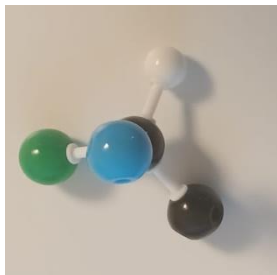


SMILES Specification Rules (4)

- Stereochemistry:
 - Chiral center: mirror images are not identical
 - @ and @@ indicate the arrangement of the four bonds attached to the chiral center
 - Consider the four bonds in the order in which they appear, left to right, in the SMILES string
 - Looking toward the central carbon from the perspective of the first bond
 - The following three bonds are counter-clockwise (@) or clockwise (@@)
 - Example:



N[C@H](Cl)(C)



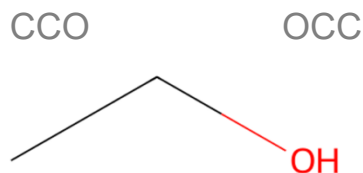
SMILES Specification Rules (5)

- Stereochemistry: trans and cis isomers
 - Arrangements of bonds typically around double bonds
 - Latin prefixes: “cis”=the side of; “trans”=the other side of
 - Can be specified by “/” and “\”
 - Two slashes are used to specify the stereochemistry:
 - Using the same slash twice indicates trans-molecule
 - Two different slashes indicate cis-molecule
 - Example:



Canonical SMILES

- SMILES strings are not unique
 - For example, they differ depending on with which molecule you start



- Canonical SMILES: generating SMILES strings such that the same molecule always produces the same SMILES
 - involves applying a standardized set of rules and algorithms

Alternative small molecule representations

- InChI (International Chemical Identifier) strings:
 - A textual identifier that provides a unique description of a chemical substance
 - Example: InChI=1S/C6H12O6/c7-1-2-3(8)4(9)5(10)6(11)12-2/h2-11H,1H2/t2-,3-,4+,5-,6-/m1/s1 is the InChI string for glucose
- Mol file format:
 - Describing chemical molecules.
 - Detailed information about the positions of atoms and the bonds connecting them:

```

3  2  0  0  0  0  0  0  0  0  0999 V2000
 21.8400 -11.9918  0.0000 C  0  0  0  0  0  0
 20.6288 -12.6940  0.0000 O  0  0  0  0  0  0
 23.0512 -12.6940  0.0000 O  0  0  0  0  0  0
 1  2  2  0  0  0
 1  3  2  0  0  0
M  END

```